4. Integral

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Definition 4.1: Simple Function. A non-negative function $\varphi : \mathbb{R} \to \mathbb{R}$ is a *simple function* if its range is a finite set of distinct non-negative real numbers $\{a_1, a_2, \ldots, a_n\}$, and the preimages

$$A_i = \varphi^{-1}(\{a_i\}) = \{x : \varphi(x) = a_i\}, \quad i = 1, 2, \ldots, n$$

are measurable sets.

The sets $A_i \in \mathcal{M}$ are pairwise disjoint and partition $\mathbb{R}$. Thus, φ has the standard representation:

$$\varphi(x) = \sum_{i=1}^{n} a_i \mathbb{I}_{A_i}(x)$$

Since each A_i is measurable, the indicator functions are measurable, making the simple function φ measurable.

Definition 4.2: Lebesgue Integral of a Simple Function. The *integral* over a measurable set $E \in \mathcal{M}$ of the simple function $\varphi = \sum_{i=1}^{n} a_i \mathbb{I}_{A_i}$ is given by

$$\int_E \varphi \, \mathrm{d}m = \sum_{i=1}^{n} a_i m(A_i \cap E).$$

By convention in measure theory, we define $0 \cdot \infty = 0$ to handle cases where $a_i = 0$ and $m(A_i \cap E) = \infty$.

Definition 4.4: Lebesgue Integral of a Non-Negative Function. For any non-negative measurable function f and $E \in \mathcal{M}$, the integral $\int_E f \, \mathrm{d}m$ is defined as

$$\int_E f \, \mathrm{d}m = \sup Y(E, f)$$

where

$$Y(E, f) = \left\{ \int_E \varphi \, \mathrm{d}m : 0 \le \varphi \le f, \varphi \text{ is a simple function} \right\}.$$

Note that the integral can evaluate to $+\infty$.

Proposition 4.5 For simple functions, Definitions 4.2 and 4.4 are equivalent.

Crux. Let $\varphi(x) = \sum_{i=1}^{n} a_i \mathbb{I}_{A_i}(x)$. Since φ is a simple function and $\varphi \leq \varphi$, we have $\int_E \varphi \,\mathrm{d}m \in Y(E,\varphi)$. Therefore, $\int_E \varphi \,\mathrm{d}m \leq \sup Y(E,\varphi)$.

For the reverse inequality, let $\psi(x) = \sum_{j=1}^{m} c_j \mathbb{I}_{F_j}(x)$ be any simple function such that $0 \leq \psi \leq \varphi$. Because A_i and F_j partition $\mathbb{R}$, we evaluate the integral on their common refinement:

$$\begin{aligned}
\int_E \psi \,\mathrm{d}m &= \sum_{i=1}^{n} \sum_{j=1}^{m} c_j m(A_i \cap F_j \cap E) \\
&\leq \sum_{i=1}^{n} \sum_{j=1}^{m} a_i m(A_i \cap F_j \cap E) \quad (\text{since } c_j \leq a_i \text{ on } A_i \cap F_j) \\
&= \sum_{i=1}^{n} a_i m(A_i \cap E) \\
&= \int_E \varphi \,\mathrm{d}m.
\end{aligned}$$

Since this holds for every $\psi \leq \varphi$, we have $\sup Y(E,\varphi) \leq \int_E \varphi \,\mathrm{d}m$. Thus, the two definitions agree. □

Theorem 4.6. *Let φ, ψ be simple functions. Then:*

(i) if $\varphi \leq \psi$, then $\int_E \varphi \,\mathrm{d}m \leq \int_E \psi \,\mathrm{d}m$,

(ii) if $A, B \in \mathcal{M}$ are disjoint sets, then

$$\int_{A\cup B} \varphi \,\mathrm{d}m = \int_A \varphi \,\mathrm{d}m + \int_B \varphi \,\mathrm{d}m,$$

(iii) for all constants $a > 0$,

$$\int_A a\varphi \,\mathrm{d}m = a \int_A \varphi \,\mathrm{d}m.$$

Crux. (i) Since $\varphi \leq \psi$ and φ is simple, $\int_E \varphi \,\mathrm{d}m \in Y(E,\psi)$. Thus, $\int_E \varphi \,\mathrm{d}m \leq \sup Y(E,\psi) = \int_E \psi \,\mathrm{d}m$.

(ii) Let $\varphi(x) = \sum_{i=1}^{n} a_i \mathbb{I}_{F_i}(x)$. By the additivity of measure m on disjoint sets:

$$\begin{aligned}
\int_{A\cup B} \varphi \,\mathrm{d}m &= \sum_{i=1}^{n} a_i m(F_i \cap (A \cup B)) \\
&= \sum_{i=1}^{n} a_i \big(m(F_i \cap A) + m(F_i \cap B)\big) \\
&= \int_A \varphi \,\mathrm{d}m + \int_B \varphi \,\mathrm{d}m.
\end{aligned}$$

(iii) Let $\varphi(x) = \sum_{i=1}^{n} c_i \mathbb{I}_{F_i}(x)$. Then:

$$\int_A a\varphi \,\mathrm{d}m = \sum_{i=1}^{n} ac_i m(F_i \cap A) = a \sum_{i=1}^{n} c_i m(F_i \cap A) = a \int_A \varphi \,\mathrm{d}m.$$

$\square$

Theorem 4.7. *Suppose f and g are non-negative measurable functions.*

(i) If $A \in \mathcal{M}$, and $f \leq g$ on A, then $\int_A f \,\mathrm{d}m \leq \int_A g \,\mathrm{d}m$.

(ii) If $B \subseteq A$, $A, B \in \mathcal{M}$, then $\int_B f \,\mathrm{d}m \leq \int_A f \,\mathrm{d}m$.

(iii) For $a \geq 0$, $\int_A af \,\mathrm{d}m = a \int_A f \,\mathrm{d}m$.

(iv) If A is null, then $\int_A f \,\mathrm{d}m = 0$.

(v) If $A, B \in \mathcal{M}$, $A \cap B = \varnothing$, then $\int_{A\cup B} f \,\mathrm{d}m = \int_A f \,\mathrm{d}m + \int_B f \,\mathrm{d}m$.

Crux. (i) Since $f \leq g$ on A, any simple function satisfying $0 \leq \varphi \leq f$ also satisfies $0 \leq \varphi \leq g$. Thus $Y(A, f) \subseteq Y(A, g)$, implying $\sup Y(A, f) \leq \sup Y(A, g)$.

(ii) From an earlier lemma, $\int_B f \,\mathrm{d}m = \int_A f\mathbb{I}_B \,\mathrm{d}m$. Since $f\mathbb{I}_B \leq f$ on A, part (i) yields $\int_A f\mathbb{I}_B \,\mathrm{d}m \leq \int_A f \,\mathrm{d}m$.

(iii) Trivial for $a = 0$. For $a > 0$, $\varphi \leq af \Leftrightarrow \frac{\varphi}{a} \leq f$. From Theorem 4.6(iii), $\int_A \varphi \,\mathrm{d}m = a \int_A \frac{\varphi}{a} \,\mathrm{d}m$, so that the entire supremum set gets scaled by a.

(iv) For any simple $0 \leq \varphi \leq f$, $\int_A \varphi \,\mathrm{d}m = \sum a_i m(E_i \cap A) = 0$ since $m(A) = 0$. Thus $Y(A, f) = \{0\}$.

(v) *Upper bound:* For any simple $0 \leq \varphi \leq f$, Theorem 4.6(ii) gives $\int_{A\cup B} \varphi \,\mathrm{d}m = \int_A \varphi \,\mathrm{d}m + \int_B \varphi \,\mathrm{d}m \leq \int_A f \,\mathrm{d}m + \int_B f \,\mathrm{d}m$. Taking the supremum yields $\int_{A\cup B} f \,\mathrm{d}m \leq \int_A f \,\mathrm{d}m + \int_B f \,\mathrm{d}m$.
Lower bound: Let $0 \leq \varphi \leq f$ on A, and $0 \leq \psi \leq f$ on B be simple functions. Define $\gamma = \varphi\mathbb{I}_A + \psi\mathbb{I}_B$, which is simple and $\gamma \leq f$ on $A \cup B$. Then $\int_A \varphi \,\mathrm{d}m + \int_B \psi \,\mathrm{d}m = \int_{A\cup B} \gamma \,\mathrm{d}m \leq \int_{A\cup B} f \,\mathrm{d}m$. Taking suprema over all such φ and ψ yields $\int_A f \,\mathrm{d}m + \int_B f \,\mathrm{d}m \leq \int_{A\cup B} f \,\mathrm{d}m$.

$\square$

Exercise 4.3 (Mean Value Theorem for the Integral) If $a \leq f(x) \leq b$ for $x \in A$, then

$$am(A) \leq \int_A f \,\mathrm{d}m \leq bm(A).$$

Crux. Since $a \leq f(x) \leq b$ on A, integrating these functions over A and applying the monotonicity property from Theorem 4.7(i) yields $\int_A a \,\mathrm{d}m \leq \int_A f \,\mathrm{d}m \leq \int_A b \,\mathrm{d}m$, from which the result immediately follows. $\square$

Theorem 4.8. *Suppose f is a non-negative measurable function. Then $f = 0$ a.e. if and only if $\int_{\mathbb{R}} f \,\mathrm{d}m = 0$.*

Crux. ($\implies$) Suppose $f = 0$ a.e., and let $N = \{x : f(x) > 0\}$ so that $m(N) = 0$. By Theorem 4.7(iv) and 4.7(v),

$$\int_{\mathbb{R}} f \,\mathrm{d}m = \int_{\mathbb{R}\setminus N} f \,\mathrm{d}m + \int_N f \,\mathrm{d}m = 0 + 0 = 0.$$

($\impliedby$) Suppose $\int_{\mathbb{R}} f \,\mathrm{d}m = 0$. Let $E_n = \{x \in \mathbb{R} : f(x) \geq 1/n\}$. Since $\frac{1}{n}\mathbb{I}_{E_n} \leq f$, monotonicity (Theorem 4.7(i)) implies

$$\frac{1}{n}m(E_n) = \int_{\mathbb{R}} \frac{1}{n}\mathbb{I}_{E_n} \,\mathrm{d}m \leq \int_{\mathbb{R}} f \,\mathrm{d}m = 0,$$

forcing $m(E_n) = 0$ for all $n \in \mathbb{N}$. The sets E_n are nested ($E_n \subseteq E_{n+1}$), and $\{x \in \mathbb{R} : f(x) > 0\} = \bigcup_{n=1}^{\infty} E_n$. By continuity of measure from below (Theorem 2.19(i)),

$$m(\{f > 0\}) = m\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} m(E_n) = 0.$$

Thus, $f = 0$ a.e. $\square$

Proposition 4.9 If f and g are non-negative measurable functions, then $f \leq g$ a.e. implies $\int_{\mathbb{R}} f \,\mathrm{d}m \leq \int_{\mathbb{R}} g \,\mathrm{d}m$.

Crux. Let $N = \{x \in \mathbb{R} : f(x) > g(x)\}$. By hypothesis, $m(N) = 0$. Since $f \leq g$ on N^c, Theorems 4.7(iv) and 4.7(i) yield:

$$\int_{\mathbb{R}} f \,\mathrm{d}m = \int_{N^c} f \,\mathrm{d}m + \int_N f \,\mathrm{d}m = \int_{N^c} f \,\mathrm{d}m \leq \int_{N^c} g \,\mathrm{d}m \leq \int_{\mathbb{R}} g \,\mathrm{d}m.$$

$\square$

Proposition 4.10 The function $f : \mathbb{R} \to \mathbb{R}$ is measurable iff both f^+ and f^- are measurable.

Crux. ($\implies$) By definition, $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. Since f and the constant function 0 are measurable, their maximums are measurable. Thus, f^+ and f^- are measurable.

($\impliedby$) We can reconstruct the original function algebraically as $f = f^+ - f^-$. Since the difference of two measurable functions is measurable, f is inherently measurable. $\square$

Theorem 4.11. *(Fatou's Lemma) If $\{f_n\}$ is a sequence of non-negative measurable functions then,*

$$\int_E \left(\liminf_{n\to\infty} f_n\right) \mathrm{d}m \le \liminf_{n\to\infty} \int_E f_n \,\mathrm{d}m.$$

Crux. Let $f \triangleq \liminf_{n\to\infty} f_n$ and $g_n \triangleq \inf_{k\ge n} f_k$. The sequence $\{g_n\}$ is monotonically increasing and $g_n \to f$. Furthermore, since $g_n \le f_k$ for all $k \ge n$, we have $\int_E g_n \,\mathrm{d}m \le \int_E f_k \,\mathrm{d}m$, which implies

$$\int_E g_n \,\mathrm{d}m \le \inf_{k\ge n} \int_E f_k \,\mathrm{d}m \implies \lim_{n\to\infty} \int_E g_n \,\mathrm{d}m \le \liminf_{n\to\infty} \int_E f_n \,\mathrm{d}m.$$

Thus, it suffices to show $\int_E f \,\mathrm{d}m \le \lim_{n\to\infty} \int_E g_n \,\mathrm{d}m$.

Let φ be any simple function such that $0 \le \varphi \le f$, and choose any constant $0 < c < 1$. Define the expanding sequence of sets $A_n \triangleq \{x \in E : g_n(x) \ge c\varphi(x)\}$. Since $g_n(x) \to f(x) \ge \varphi(x) > c\varphi(x)$ whenever $\varphi(x) > 0$, the sets A_n expand to cover the support of φ, so $\bigcup_{n=1}^{\infty} A_n = E$ (up to a null set).

By monotonicity,

$$\int_E g_n \,\mathrm{d}m \ge \int_{A_n} g_n \,\mathrm{d}m \ge c \int_{A_n} \varphi \,\mathrm{d}m.$$

Taking the limit as $n \to \infty$ and applying continuity of measure to the right side yields

$$\lim_{n\to\infty} \int_E g_n \,\mathrm{d}m \ge c \int_E \varphi \,\mathrm{d}m.$$

Since this holds for any $0 < c < 1$, we can let $c \uparrow 1$ to obtain $\lim_{n\to\infty} \int_E g_n \,\mathrm{d}m \ge \int_E \varphi \,\mathrm{d}m$. Finally, taking the supremum over all such simple functions $\varphi \le f$ gives $\lim_{n\to\infty} \int_E g_n \,\mathrm{d}m \ge \int_E f \,\mathrm{d}m$, completing the proof. $\square$

Theorem 4.13. *(Monotone Convergence Theorem) If $\{f_n\}$ is a sequence of non-negative measurable functions, and $f_n \uparrow f$ pointwise on E, then*

$$\lim_{n\to\infty} \int_E f_n \,\mathrm{d}m = \int_E f \,\mathrm{d}m.$$

Crux. Since $f_n \to f$ pointwise, Fatou's Lemma provides the lower bound:

$$\int_E f \,\mathrm{d}m \le \liminf_{n\to\infty} \int_E f_n \,\mathrm{d}m.$$

Conversely, since $f_n \uparrow f$, we have $f_n \le f$ for all n. By monotonicity (Proposition 4.9), $\int_E f_n \,\mathrm{d}m \le \int_E f \,\mathrm{d}m$, which gives the upper bound:

$$\limsup_{n\to\infty} \int_E f_n \,\mathrm{d}m \le \int_E f \,\mathrm{d}m.$$

Combining these bounds yields $\limsup \int_E f_n \,\mathrm{d}m \le \int_E f \,\mathrm{d}m \le \liminf \int_E f_n \,\mathrm{d}m$, forcing the limit to exist and equal $\int_E f \,\mathrm{d}m$. $\square$

Corollary 4.14 Suppose $\{f_n\}$ and f are non-negative measurable functions. If $f_n \nearrow f$ almost everywhere on E, then $\lim_{n\to\infty} \int_E f_n \,\mathrm{d}m = \int_E f \,\mathrm{d}m$.

Crux. Let $N = \{x \in E : f_n(x) \not\nearrow f(x)\}$. By hypothesis, $m(N) = 0$. Since $f_n \nearrow f$ pointwise on $E \setminus N$, and as integrals vanish on N, applying the Monotone Convergence Theorem on $E \setminus N$ yields:

$$\int_E f_n \,\mathrm{d}m = \int_{E\setminus N} f_n \,\mathrm{d}m = \int_{E\setminus N} f_n \,\mathrm{d}m \nearrow \int_{E\setminus N} f \,\mathrm{d}m = \int_E f \,\mathrm{d}m.$$

$\square$

Proposition 4.15 For any non-negative measurable function f, there exists a sequence of non-negative simple functions $\{s_n\}$ such that $s_n \nearrow f$ pointwise.

Crux. For each $n \geq 1$, partition $[0, n)$ into $n2^n$ intervals of width 2^{-n}. Define the measurable sets:

$$E_{n,i} = \{x : i2^{-n} \leq f(x) < (i+1)2^{-n}\} \quad \text{for } 0 \leq i < n2^n, \quad \text{and} \quad E_n = \{x : f(x) \geq n\}.$$

Construct the simple function by taking the lower bound on each set:

$$s_n(x) = \sum_{i=0}^{n2^n-1} i2^{-n}\mathbb{I}_{E_{n,i}}(x) + n\mathbb{I}_{E_n}(x).$$

Moving from n to $n+1$ simply bisects the intervals and raises the height cap, forcing monotonicity: $s_n \leq s_{n+1} \leq f$.

For pointwise convergence: if $f(x) < \infty$, then for sufficiently large n, $f(x) < n$, which bounds the approximation error by the interval width: $0 \leq f(x) - s_n(x) < 2^{-n} \to 0$. If $f(x) = \infty$, $s_n(x) = n \to \infty$. Thus, $s_n \nearrow f$. $\square$

Definition 4.16 If $E \in \mathcal{M}$ and the measurable function f has both $\int_E f^+ \,\mathrm{d}m$ and $\int_E f^- \,\mathrm{d}m$ finite, then we say that f is *integrable,* and define

$$\int_E f \,\mathrm{d}m \triangleq \int_E f^+ \,\mathrm{d}m - \int_E f^- \,\mathrm{d}m.$$

The set of all functions that are integrable over E is denoted by $\mathcal{L}^1(E)$ (we drop (E) if it is understood from the context).

Definition 4.16.33 A function $f : I \to \mathbb{R}$, where $I \subseteq \mathbb{R}$, is called locally integrable if its Lebesgue integral over every compact subset of I (which means every closed and bounded interval $[a, b] \subset I$) is finite.

Lemma 4.16.66. *Let $f : I \to \mathbb{R}$ be a bounded, locally Lebesgue integrable function on an interval $I \subseteq \mathbb{R}$. Then its indefinite Lebesgue integral, defined as $F(x) = \int_{[a,x]} f(t)\,\mathrm{d}m(t)$ for a fixed $a \in I$, is Lipschitz continuous (and therefore continuous) everywhere on I.*

Crux. Since f is bounded, there exists a finite constant $M > 0$ such that $|f(t)| \leq M$ for almost all $t \in I$. Let $x, y \in I$, and assume without loss of generality that $x < y$. Evaluating the absolute difference of the indefinite integral yields:

$$\begin{aligned}
|F(y) - F(x)| &= \left| \int_{[a,y]} f(t)\,\mathrm{d}m(t) - \int_{[a,x]} f(t)\,\mathrm{d}m(t) \right| \\
&= \left| \int_{[x,y]} f(t)\,\mathrm{d}m(t) \right| \\
&\leq \int_{[x,y]} |f(t)|\,\mathrm{d}m(t) \quad \text{(Integral Triangle Inequality)} \\
&\leq \int_{[x,y]} M\,\mathrm{d}m(t) \\
&= M(y - x) \\
&= M\,|y - x|\,.
\end{aligned}$$

This establishes the bound $|F(y) - F(x)| \leq M\,|y - x|$, which is the definition of Lipschitz continuity. By the squeeze theorem, taking the limit as $y \to x$ forces $|F(y) - F(x)| \to 0$. Thus, F is continuous everywhere on I. □

Proposition 4.17 If f and g are integrable and $f \leq g$, then $\int f\,\mathrm{d}m \leq \int g\,\mathrm{d}m$.

Crux. Since $f \leq g$, we can write $f^+ - f^- \leq g^+ - g^-$. Rearranging to keep all terms strictly non-negative yields:

$$f^+ + g^- \leq g^+ + f^-.$$

Because all four functions are non-negative and measurable, we can integrate both sides using additivity and monotonicity:

$$\int f^+\,\mathrm{d}m + \int g^-\,\mathrm{d}m \leq \int g^+\,\mathrm{d}m + \int f^-\,\mathrm{d}m.$$

Since f and g are integrable, all of these integrals are finite. Therefore, we can safely subtract $\int f^-\,\mathrm{d}m$ and $\int g^-\,\mathrm{d}m$ from both sides to obtain:

$$\int f^+\,\mathrm{d}m - \int f^-\,\mathrm{d}m \leq \int g^+\,\mathrm{d}m - \int g^-\,\mathrm{d}m \implies \int f\,\mathrm{d}m \leq \int g\,\mathrm{d}m.$$

□

Note: Strict Inequality. If $f \leq g$ and $f < g$ on a set A of positive measure ($m(A) > 0$), then the integral inequality is strict: $\int f \, \mathrm{d}m < \int g \, \mathrm{d}m$.

Crux. Let $Z = (g - f)\mathbb{I}_A$. Define the expanding sequence of sets $A_k = \{Z \geq 1/k\}$. Because $A = \bigcup_{k=1}^{\infty} A_k$ and $m(A) > 0$, continuity of measure from below guarantees there exists some finite k where $m(A_k) > 0$. Bounding the integral on this subset strictly separates it from zero:

$$\int Z \, \mathrm{d}m \geq \int_{A_k} Z \, \mathrm{d}m \geq \int_{A_k} \frac{1}{k} \, \mathrm{d}m = \frac{1}{k} m(A_k) > 0$$

□

Theorem 4.19. *(Linearity) For any integrable functions f, g, their sum $f + g$ is also integrable and*

$$\int_E (f + g) \, \mathrm{d}m = \int_E f \, \mathrm{d}m + \int_E g \, \mathrm{d}m.$$

Crux. We proceed in three standard steps:

- **Non-negative simple functions:** Let $f = \sum a_i \mathbb{I}_{A_i}$ and $g = \sum b_j \mathbb{I}_{B_j}$. Their sum is $\sum_{i,j} (a_i + b_j) \mathbb{I}_{A_i \cap B_j}$. By the finite additivity of measure over the disjoint partition $A_i \cap B_j$, the integral distributes linearly: $\int_E (f + g) \, \mathrm{d}m = \int_E f \, \mathrm{d}m + \int_E g \, \mathrm{d}m$.

- **Non-negative measurable functions:** By Proposition 4.15, there exist simple sequences $s_n \nearrow f$ and $t_n \nearrow g$, meaning $(s_n + t_n) \nearrow (f + g)$. Using the previous step and the Monotone Convergence Theorem:

$$\int_E (f+g) \, \mathrm{d}m = \lim_{n\to\infty} \int_E (s_n + t_n) \, \mathrm{d}m = \lim_{n\to\infty} \int_E s_n \, \mathrm{d}m + \lim_{n\to\infty} \int_E t_n \, \mathrm{d}m = \int_E f \, \mathrm{d}m + \int_E g \, \mathrm{d}m.$$

- **Arbitrary integrable functions:** Integrability of $f + g$ follows from $|f + g| \leq |f| + |g|$. Since $f + g = f^+ - f^- + g^+ - g^-$, we can rearrange into strictly non-negative terms:

  $$(f + g)^+ + f^- + g^- = (f + g)^- + f^+ + g^+.$$

  Integrating both sides (valid by the previous step) and rearranging the finite integrals yields $\int_E (f + g) \, \mathrm{d}m = \int_E f \, \mathrm{d}m + \int_E g \, \mathrm{d}m$.

□

Proposition 4.20 If f is integrable and $c \in \mathbb{R}$, then

$$\int_E (cf) \, \mathrm{d}m = c \int_E f \, \mathrm{d}m.$$

Crux. For $c > 0$ and $f \geq 0$, $\int_E cf \, \mathrm{d}m = c \int_E f \, \mathrm{d}m$ holds by the scaling property of non-negative integrals (Theorem 4.7). For arbitrary integrable $f = f^+ - f^-$ and $c > 0$, we have $(cf)^+ = cf^+$ and $(cf)^- = cf^-$. By linearity:

$$\int_E cf \, \mathrm{d}m = \int_E cf^+ \, \mathrm{d}m - \int_E cf^- \, \mathrm{d}m = c \int_E f^+ \, \mathrm{d}m - c \int_E f^- \, \mathrm{d}m = c \int_E f \, \mathrm{d}m.$$

If $c < 0$, the signs flip: $(cf)^+ = |c| f^-$ and $(cf)^- = |c| f^+$. Applying the positive constant case yields:

$$\begin{aligned} \int_E cf \, \mathrm{d}m &= \int_E |c| f^- \, \mathrm{d}m - \int_E |c| f^+ \, \mathrm{d}m = |c| \int_E f^- \, \mathrm{d}m - |c| \int_E f^+ \, \mathrm{d}m \\ &= -|c| \left(\int_E f^+ \, \mathrm{d}m - \int_E f^- \, \mathrm{d}m \right) = c \int_E f \, \mathrm{d}m. \end{aligned}$$

□

Proposition 4.22 Let f, g be integrable.

(i) If $\int_A f \, \mathrm{d}m \leq \int_A g \, \mathrm{d}m$ for all measurable $A \subseteq E$, then $f \leq g$ a.e.

(ii) If $\int_A f \, \mathrm{d}m = \int_A g \, \mathrm{d}m$ for all measurable $A \subseteq E$, then $f = g$ a.e.

Crux. (i) Let $h = g - f$. We are given $\int_A h \, \mathrm{d}m \geq 0$ for all A. Define $A_n = \{x \in E : h(x) \leq -1/n\}$. Then:

$$\int_{A_n} h \, \mathrm{d}m \leq -\frac{1}{n} m(A_n).$$

Since $\int_{A_n} h \, \mathrm{d}m \geq 0$, it must be that $m(A_n) = 0$ for all n. The set where $h < 0$ is $A = \bigcup_{n=1}^{\infty} A_n$, a countable union of null sets. Hence A is null and $h \geq 0$ a.e.

(ii) Applying the above result twice, the sets on which $f > g$ and the sets on which $f < g$ are null, forcing $f = g$ a.e.

□

Proposition 4.23

(i) An integrable function is finite a.e.

(ii) For a measurable function f and measurable set A, $m(A) \inf_A f \leq \int_A f \, \mathrm{d}m \leq m(A) \sup_A f$.

(iii) $\left| \int f \, \mathrm{d}m \right| \leq \int |f| \, \mathrm{d}m$.

(iv) If $f \geq 0$ and $\int f \, \mathrm{d}m = 0$, then $f = 0$ a.e.

Crux. (i) Let $A = \{x : |f(x)| = \infty\}$. Since f is integrable, $\int |f| \, \mathrm{d}m < \infty$. For any n, $|f| \geq n\mathbb{I}_A$, so monotonicity yields $n \cdot m(A) \leq \int |f| \, \mathrm{d}m < \infty$. Letting $n \to \infty$ forces $m(A) = 0$.

(ii) Since $\inf_A f \leq f \leq \sup_A f$ pointwise on A, integrating over A directly yields the result.

(iii) The pointwise inequality $-|f| \leq f \leq |f|$ integrates to $-\int |f| \, \mathrm{d}m \leq \int f \, \mathrm{d}m \leq \int |f| \, \mathrm{d}m$, which is equivalent to $\left|\int f \, \mathrm{d}m\right| \leq \int |f| \, \mathrm{d}m$.

(iv) Let $E_n = \{x : f(x) \geq 1/n\}$. Since $f \geq \frac{1}{n}\mathbb{I}_{E_n}$, we have $0 = \int f \, \mathrm{d}m \geq \int_{E_n} f \, \mathrm{d}m \geq \frac{1}{n}m(E_n)$, forcing $m(E_n) = 0$ for all n. The set $\{x \in \mathbb{R} : f(x) > 0\} = \bigcup_{n=1}^{\infty} E_n$, which is a countable union of null sets and therefore null. Thus $f = 0$ a.e.

$\square$

Theorem 4.24. *Let $f \geq 0$ be measurable. Then the set function $\mu(A) = \int_A f \, \mathrm{d}m$ defines a measure on $\mathcal{M}$.*

Crux. Since $\int_\emptyset f \, \mathrm{d}m = 0$, $\mu(\emptyset) = 0$. We must show countable additivity. Let $\{E_i\}_{i=1}^{\infty}$ be a disjoint sequence of measurable sets, and $E = \bigcup_{i=1}^{\infty} E_i$. Define the sequence of functions $S_n = \sum_{i=1}^{n} f\mathbb{I}_{E_i}$. Since $f \geq 0$, the sequence is increasing and $S_n \nearrow f\mathbb{I}_E$ pointwise. By linearity, $\int S_n \, \mathrm{d}m = \sum_{i=1}^{n} \int_{E_i} f \, \mathrm{d}m$. Applying the Monotone Convergence Theorem to S_n yields:

$$\mu(E) = \int f\mathbb{I}_E \, \mathrm{d}m = \lim_{n\to\infty} \int S_n \, \mathrm{d}m = \lim_{n\to\infty} \sum_{i=1}^{n} \int_{E_i} f \, \mathrm{d}m = \sum_{i=1}^{\infty} \mu(E_i).$$

Therefore, μ is countably additive and is a measure. $\square$

Theorem 4.26. *(Dominated Convergence Theorem) Suppose $E \in \mathcal{M}$. Let $\{f_n\}$ be a sequence of measurable functions such that $|f_n| \leq g$ a.e. on E for all n, where g is integrable over E. If $f_n \to f$ pointwise a.e., then f is integrable over E and*

$$\lim_{n\to\infty} \int_E f_n \, \mathrm{d}m = \int_E f \, \mathrm{d}m.$$

Crux. Since $|f_n| \leq g$ a.e., taking the limit gives $|f| \leq g$ a.e., so f is integrable. Furthermore, both $g + f_n \geq 0$ and $g - f_n \geq 0$.

Applying Fatou's Lemma to the non-negative sequence $g + f_n$:

$$\int_E (g + f) \, \mathrm{d}m \leq \liminf_{n\to\infty} \int_E (g + f_n) \, \mathrm{d}m = \int_E g \, \mathrm{d}m + \liminf_{n\to\infty} \int_E f_n \, \mathrm{d}m.$$

Since g is integrable, $\int_E g \, \mathrm{d}m < \infty$. Subtracting it from both sides yields $\int_E f \, \mathrm{d}m \leq \liminf_{n\to\infty} \int_E f_n \, \mathrm{d}m$.

Applying Fatou's Lemma to the non-negative sequence $g - f_n$ (and using $\liminf(-x_n) = -\limsup x_n$):

$$\int_E (g-f)\,\mathrm{d}m \le \liminf_{n\to\infty} \int_E (g-f_n)\,\mathrm{d}m = \int_E g\,\mathrm{d}m - \limsup_{n\to\infty} \int_E f_n\,\mathrm{d}m.$$

Subtracting $\int_E g\,\mathrm{d}m$ yields $\limsup_{n\to\infty} \int_E f_n\,\mathrm{d}m \le \int_E f\,\mathrm{d}m$.

Combining the bounds, we get $\limsup \int_E f_n\,\mathrm{d}m \le \int_E f\,\mathrm{d}m \le \liminf \int_E f_n\,\mathrm{d}m$, which forces the limit to exist and equal $\int_E f\,\mathrm{d}m$. $\square$

Proposition 4.28 Suppose f is integrable and define $g_n = f\mathbb{I}_{[-n,n]}$ and $h_n = \min(f, n)$. Then $\lim_{n\to\infty} \int |f - g_n|\,\mathrm{d}m = 0$ and $\lim_{n\to\infty} \int |f - h_n|\,\mathrm{d}m = 0$.

Crux. As $n \to \infty$, both $g_n \to f$ and $h_n \to f$ pointwise, meaning the difference sequences $|f - g_n| \to 0$ and $|f - h_n| \to 0$ pointwise. By definition, $|g_n| \le |f|$ and $|h_n| \le |f|$ for all n. The triangle inequality allows us to bound both difference sequences by the integrable function $2|f|$:

$$|f - g_n| \le |f| + |g_n| \le 2|f| \quad \text{and} \quad |f - h_n| \le |f| + |h_n| \le 2|f|.$$

Since f is integrable, $2|f|$ is integrable. Applying the Dominated Convergence Theorem directly to the non-negative sequences $|f - g_n|$ and $|f - h_n|$ shows their integrals converge to $\int 0\,\mathrm{d}m = 0$. $\square$

Proposition 4.29 For a sequence of non-negative measurable functions $\{f_n\}$,

$$\int \sum_{n=1}^{\infty} f_n\,\mathrm{d}m = \sum_{n=1}^{\infty} \int f_n\,\mathrm{d}m.$$

Crux. Define the partial sums $g_k \triangleq \sum_{n=1}^{k} f_n$. Since $f_n \ge 0$, the sequence $\{g_k\}$ is monotonically increasing with $g_k \nearrow \sum_{n=1}^{\infty} f_n$ pointwise. By linearity, the integral of the finite sum is $\int g_k\,\mathrm{d}m = \sum_{n=1}^{k} \int f_n\,\mathrm{d}m$. Applying the Monotone Convergence Theorem to the sequence $\{g_k\}$ yields the desired result. $\square$

Theorem 4.30. *(Beppo-Levi)*

Suppose that $\sum_{k=1}^{\infty} \int |f_k|\,\mathrm{d}m < \infty$. Then the series $\sum_{k=1}^{\infty} f_k$ converges for almost all x, its sum is integrable, and

$$\int \sum_{k=1}^{\infty} f_k\,\mathrm{d}m = \sum_{k=1}^{\infty} \int f_k\,\mathrm{d}m.$$

Crux. By Proposition 4.29 applied to the non-negative sequence $|f_k|$, we have $\int \sum_{k=1}^{\infty} |f_k|\,\mathrm{d}m = \sum_{k=1}^{\infty} \int |f_k|\,\mathrm{d}m < \infty$. Since its integral is finite, the function $\sum_{k=1}^{\infty} |f_k|$ is finite a.e. (Proposition 4.23(i)). Absolute convergence implies that $\sum_{k=1}^{\infty} f_k$ converges a.e., and its limit is bounded by the integrable sum, making it integrable.

Define the partial sums $g_n = \sum_{k=1}^n f_k$. We have $g_n \to \sum_{k=1}^\infty f_k$ a.e., and $|g_n| \le \sum_{k=1}^\infty |f_k|$ for all n. Because the dominating function is integrable, applying the Dominated Convergence Theorem yields:

$$\int \sum_{k=1}^{\infty} f_k \,\mathrm{d}m = \lim_{n\to\infty} \int g_n \,\mathrm{d}m = \lim_{n\to\infty} \sum_{k=1}^{n} \int f_k \,\mathrm{d}m = \sum_{k=1}^{\infty} \int f_k \,\mathrm{d}m.$$

$\square$

Proposition 4.32 If $f : [a,b] \to \mathbb{R}$ is continuous, then f is integrable and the function $F(x) = \int_{[a,x]} f \,\mathrm{d}m$ is differentiable on (a,b) with derivative $F'(x) = f(x)$.

Crux. For $h > 0$, the domain splits into disjoint intervals, so $F(x+h) - F(x) = \int_{[x,x+h]} f \,\mathrm{d}m$. By Proposition 4.23(ii), bounding the integral by the infimum m_h and supremum M_h of f on $[x, x+h]$ gives:

$$m_h \cdot h \le \int_{[x,x+h]} f \,\mathrm{d}m \le M_h \cdot h.$$

Dividing by h yields

$$m_h \le \frac{F(x+h) - F(x)}{h} \le M_h.$$

Because f is continuous at x, as $h \to 0$, both the infimum m_h and supremum M_h converge to $f(x)$. By the Squeeze Theorem, the limit is $f(x)$, so $F'(x) = f(x)$. (The symmetric argument holds for $h < 0$). $\square$

Theorem 4.33. *Let $f : [a,b] \to \mathbb{R}$ be bounded.*

(i) f is Riemann-integrable if and only if f is a.e. continuous with respect to Lebesgue measure on $[a,b]$.

(ii) Riemann-integrable functions on $[a,b]$ are integrable with respect to Lebesgue measure on $[a,b]$ and the integrals are the same.

Crux. Take a sequence of refining partitions of $E = [a,b]$ with mesh size $\to 0$. Define the lower and upper step functions $l_n = \sum m_i \mathbb{I}_{I_i}$ and $u_n = \sum M_i \mathbb{I}_{I_i}$. As the partitions refine, the bounded sequences converge monotonically: $l_n \nearrow l$ and $u_n \searrow u$, with $l \le f \le u$ everywhere. Crucially, $l(x) = u(x)$ if and only if f is continuous at x.

Since f is bounded on a set of finite measure, the Dominated Convergence Theorem guarantees $\int_E l_n \,\mathrm{d}m \to \int_E l \,\mathrm{d}m$ and $\int_E u_n \,\mathrm{d}m \to \int_E u \,\mathrm{d}m$.

(i): f is Riemann integrable $\iff$ the upper and lower Darboux limits match $\iff$ $\int_E u \,\mathrm{d}m = \int_E l \,\mathrm{d}m \iff \int_E (u - l) \,\mathrm{d}m = 0$. Since $u \ge l$, (apply Theorem 4.24 with $u - l \ge 0$) this holds $\iff$ $u = l$ a.e., which means f is continuous a.e.

(ii): If f is Riemann integrable, then $l = f = u$ a.e. Since u is the limit of measurable simple functions, it is Lebesgue measurable. Because $f = u$ almost everywhere on a complete measure space (Theorem 3.12), f is Lebesgue measurable, and its Lebesgue integral equals its Riemann integral $\int_E u \, \mathrm{d}m$. □

Definition: Improper Riemann Integral The improper Riemann integral of a function f over $\mathbb{R}$ is defined as:

$$\int_{-\infty}^{\infty} f(x) \, \mathrm{d}x \triangleq \lim_{a \to -\infty, b \to \infty} \int_a^b f(x) \, \mathrm{d}x$$

provided the double limit exists.

Theorem 4.36. *If $f \geq 0$ and the improper Riemann integral of f exists, then f is Lebesgue integrable and $\int_{\mathbb{R}} f \, \mathrm{d}m = \int_{-\infty}^{\infty} f(x) \, \mathrm{d}x$.*

Crux. Define $f_n = f \mathbb{I}_{[-n,n]}$. Since $f \geq 0$, the sequence is non-negative and monotonically increasing to f ($f_n \nearrow f$). Because the improper Riemann integral exists, f is Riemann integrable on every bounded interval $[-n, n]$. By the preceding equivalence theorem on bounded intervals:

$$\int_{\mathbb{R}} f_n \, \mathrm{d}m = \int_{[-n,n]} f \, \mathrm{d}m = \int_{-n}^{n} f(x) \, \mathrm{d}x.$$

Applying the Monotone Convergence Theorem (MCT) to f_n:

$$\int_{\mathbb{R}} f \, \mathrm{d}m = \lim_{n \to \infty} \int_{\mathbb{R}} f_n \, \mathrm{d}m = \lim_{n \to \infty} \int_{-n}^{n} f(x) \, \mathrm{d}x = \int_{-\infty}^{\infty} f(x) \, \mathrm{d}x.$$

Since the improper Riemann integral is finite by hypothesis, $\int_{\mathbb{R}} f \, \mathrm{d}m < \infty$, proving f is Lebesgue integrable. □

Theorem 4.38. *If f is a bounded measurable function on $[a, b]$ and $\epsilon > 0$, there exists a step function h such that $\int_{[a,b]} |f - h| \, \mathrm{d}m < \epsilon$.*

Crux. Assume $0 \leq f \leq M$. Since f is bounded and measurable, there exists a simple function $\varphi = \sum_{i=1}^{n} a_i \mathbb{I}_{A_i}$ such that $0 \leq \varphi \leq f$ and $\int_{[a,b]} |f - \varphi| \, \mathrm{d}m < \epsilon/2$.

By outer regularity, each measurable A_i can be covered by an open set $O_i \supset A_i$ with $m(O_i \setminus A_i) < \frac{\epsilon}{4Mn}$. Since O_i is a countable union of disjoint open intervals, we can truncate this to a finite union of intervals $G_i \subset O_i$ such that $m(O_i \setminus G_i) < \frac{\epsilon}{4Mn}$. The indicator $\mathbb{I}_{G_i}$ is a step function. Defining $h = \sum_{i=1}^{n} a_i \mathbb{I}_{G_i}$, we bound the error using the symmetric difference $A_i \Delta G_i$:

$$\int_{[a,b]} |\varphi - h| \, \mathrm{d}m \leq \sum_{i=1}^{n} M \cdot m(A_i \Delta G_i) \leq \sum_{i=1}^{n} M \left(m(O_i \setminus A_i) + m(O_i \setminus G_i) \right) < \sum_{i=1}^{n} M \left(\frac{\epsilon}{2Mn} \right) = \frac{\epsilon}{2}.$$

By the triangle inequality, $\int_{[a,b]} |f - h| \, \mathrm{d}m \leq \int_{[a,b]} |f - \varphi| \, \mathrm{d}m + \int_{[a,b]} |\varphi - h| \, \mathrm{d}m < \epsilon$. For arbitrary bounded measurable f, apply this construction separately to f^+ and f^-. □

Theorem 4.39. *(Continuous functions are dense in $\mathcal{L}^1$)*

Given $f \in \mathcal{L}^1$ and $\epsilon > 0$, we can find a continuous function g, vanishing outside some finite interval, such that $\int |f - g| \,\mathrm{d}m < \epsilon$.

Crux. **Step 1: Truncation.** By Proposition 4.28, we can simultaneously bound f and restrict its domain. For large enough N, the bounded function $f_N = \max(\min(f, N), -N) \cdot \mathbb{I}_{[-N,N]}$ satisfies $\int |f - f_N| \,\mathrm{d}m < \epsilon/3$.

Step 2: Step Function. Since f_N is bounded and supported on $[-N, N]$, Theorem 4.38 guarantees a step function h on $[-N, N]$ such that $\int |f_N - h| \,\mathrm{d}m < \epsilon/3$.

Step 3: Continuous Interpolation. The step function h consists of finitely many jumps. We can construct a continuous function g by replacing these vertical jumps with steep, continuous linear ramps (and tapering g to 0 just outside $[-N, N]$). Because h is bounded by some M, modifying h on a set of total measure δ creates an integral error at most $2M\delta$. By making the ramps sufficiently steep (δ small), we ensure $\int |h - g| \,\mathrm{d}m < \epsilon/3$.

Conclusion: By the triangle inequality,

$$\int |f - g| \,\mathrm{d}m \le \int |f - f_N| \,\mathrm{d}m + \int |f_N - h| \,\mathrm{d}m + \int |h - g| \,\mathrm{d}m < \epsilon.$$

$\square$

Lemma 4.40 (Riemann-Lebesgue) Suppose $f \in \mathcal{L}^1$. Then the sequences $s_k = \int f(x) \sin(kx) \,\mathrm{d}m$ and $c_k = \int f(x) \cos(kx) \,\mathrm{d}m$ both converge to 0 as $k \to \infty$.

Crux. Using the substitution $y = x - \pi/k$, we get $s_k = -\int f(x + \pi/k) \sin(kx) \,\mathrm{d}m$. Averaging this with the original integral gives:

$$2|s_k| = \left| \int (f(x) - f(x + \pi/k)) \sin(kx) \,\mathrm{d}m \right| \le \int |f(x + \pi/k) - f(x)| \,\mathrm{d}m.$$

Let $\delta = \pi/k$. We must show the

"Translation Continuity": $\int |f(x + \delta) - f(x)| \,\mathrm{d}m \to 0$ as $\delta \to 0$. (In general, if $f \in \mathcal{L}^p(\mathbb{R}^n)$, then the translation $f_h(x) \triangleq f(x - h)$ satisfies $\lim_{h\to 0} \|f_h - f\|_p = 0$).

By Theorem 4.39, choose a continuous function g vanishing outside an interval of length w such that $\int |f - g| \,\mathrm{d}m < \epsilon/3$. Translation invariance of the Lebesgue measure ensures $\int |f(x + \delta) - g(x + \delta)| \,\mathrm{d}m < \epsilon/3$.

Because g is continuous and compactly supported, it is uniformly continuous. Since we are considering the case where $\delta \to 0$, we can assume that $\delta < 1$, so that the support of the shifted difference is at most $w + 1$. For small enough δ, $|g(x + \delta) - g(x)| < \frac{\epsilon}{3(w+1)}$ for all x. Integrating gives $\int |g(x + \delta) - g(x)| \,\mathrm{d}m < \epsilon/3$.

By the triangle inequality:

$$\int |f(x+\delta) - f(x)| \,\mathrm{d}m \le \int |f(x+\delta) - g(x+\delta)| \,\mathrm{d}m + \int |g(x+\delta) - g(x)| \,\mathrm{d}m + \int |g(x) - f(x)| \,\mathrm{d}m < \epsilon.$$

Thus $s_k \to 0$. The proof for c_k is identical. □

Theorem 4.41. *Given a random variable $X : \Omega \to \mathbb{R}$ and a Borel measurable function g,*

$$\int_\Omega g(X) \,\mathrm{d}P = \int_\mathbb{R} g \,\mathrm{d}P_X.$$

ERRATA: The text omits a critical technical assumption here. The function g must be **Borel measurable**, not just Lebesgue measurable. If g is not Borel measurable, the composition $g(X)$ is not guaranteed to be a valid random variable on Ω. Without this assumption, the integral on the LHS is not well-defined.

Crux. We apply the standard measure theory bootstrapping machine:

(1) **Indicators:** If $g = \mathbb{I}_A$ for a Borel set A, the LHS is $\int_\Omega \mathbb{I}_{\{X \in A\}} \,\mathrm{d}P = P(X^{-1}(A))$. The RHS is $\int_\mathbb{R} \mathbb{I}_A \,\mathrm{d}P_X = P_X(A)$. By the definition of the induced measure P_X, these are identically equal.

(2) **Simple Functions:** By linearity of the integral, the result holds for finite linear combinations of indicators ($g = \sum a_i \mathbb{I}_{A_i}$).

(3) **Non-negative Functions:** For Borel $g \ge 0$, choose simple functions $s_n \nearrow g$. By the almighty Monotone Convergence Theorem (MCT) applied to both sides, the limits are equal.

(4) **General Functions:** For arbitrary integrable g, split $g = g^+ - g^-$. Apply step 3 to the non-negative components and subtract via linearity.

□

Theorem 4.42. *If $P_X = \sum_i p_i P_i$, where the P_i are probability measures, $\sum_i p_i = 1$, and $p_i \ge 0$, then*

$$\int g \,\mathrm{d}P_X = \sum_i p_i \int g \,\mathrm{d}P_i.$$

Crux. We apply the standard measure-theoretic machine:

(1) **Indicators:** If $g = \mathbb{I}_A$ for a Borel set A, the LHS is $P_X(A)$ and the RHS is $\sum_i p_i P_i(A)$. These are equal by the hypothesis definition of P_X.

(2) **Simple Functions:** By linearity of the integral, the equality extends to finite linear combinations of indicators ($g = \sum a_k \mathbb{I}_{A_k}$).

(3) **Non-negative Functions:** For Borel $g \geq 0$, choose simple functions $s_n \nearrow g$. To legally pass the limit $n \to \infty$ through the infinite sum $\sum_i$ on the RHS, define the non-negative increments $f_1 = s_1$ and $f_k = s_k - s_{k-1}$ for $k \geq 2$. Thus, $s_n = \sum_{k=1}^{n} f_k$ and $g = \sum_{k=1}^{\infty} f_k$. By Proposition 4.29, the integral of this infinite sum is the sum of integrals. Because all terms $p_i \int f_k \, \mathrm{d}P_i$ are non-negative, we can unconditionally swap the order of summation ($\sum_i \sum_k = \sum_k \sum_i$), which reconstructs the limit $\lim_{n\to\infty} \sum_i p_i \int s_n \, \mathrm{d}P_i$. Equating this with the MCT limit on the LHS yields the result.

(4) **General Functions:** For arbitrary integrable g, decompose $g = g^+ - g^-$, apply Step 3 to each part, and subtract via linearity.

□

Definition: Absolutely Continuous Measures A measure P of the form

$$A \mapsto P(A) = \int_A f \, \mathrm{d}m$$

with a non-negative integrable function f is called *absolutely continuous* (with respect to Lebesgue measure). The function f is called the *density* of P. If $\int_\Omega f \, \mathrm{d}m = 1$ (equivalently $P(\Omega) = 1$), then P is a probability measure.

Definition: Cumulative Distribution Function The *cumulative distribution function* (CDF) corresponding to a density is:

$$F(y) \triangleq \int_{-\infty}^{y} f(x) \, \mathrm{d}x.$$

By Lebesgue's version of the Fundamental Theorem of Calculus, $F'(x) = f(x)$ almost everywhere.

More generally, a random variable $X : \Omega \to \mathbb{R}$ on a probability space $(\Omega, \mathcal{F}, P)$ has a CDF defined without needing a density:

$$F_X(y) \triangleq P(\{\omega \in \Omega : X(\omega) \leq y\}) = P_X((-\infty, y]).$$

Proposition 4.44.

(i) *F_X is non-decreasing ($y_1 \leq y_2 \implies F_X(y_1) \leq F_X(y_2)$).*

(ii) $\lim_{y\to\infty} F_X(y) = 1$ *and* $\lim_{y\to-\infty} F_X(y) = 0$.

(iii) *F_X is right-continuous ($\lim_{y\to y_0^+} F_X(y) = F_X(y_0)$).*

Crux. (i) Because $y_1 \le y_2 \implies (-\infty, y_1] \subseteq (-\infty, y_2]$, monotonicity of the measure P_X gives $F_X(y_1) \le F_X(y_2)$.

(ii) Let $B_n = (-\infty, n]$ and $A_n = (-\infty, -n]$. The sequence $\{B_n\}$ increases to $\mathbb{R}$, and $\{A_n\}$ decreases to $\varnothing$. By continuity of probability measure (Theorem 2.19):

$$\lim_{y\to\infty} F_X(y) = \lim_{n\to\infty} P_X(B_n) = P_X\left(\bigcup_{n=1}^{\infty} B_n\right) = P_X(\mathbb{R}) = 1$$

$$\lim_{y\to-\infty} F_X(y) = \lim_{n\to\infty} P_X(A_n) = P_X\left(\bigcap_{n=1}^{\infty} A_n\right) = P_X(\varnothing) = 0$$

(iii) For right-continuity at y_0, let $C_n = (-\infty, y_0 + 1/n]$. The sequence $\{C_n\}$ decreases to $(-\infty, y_0]$. Since P_X is a finite measure, Theorem 2.19(ii) applies:

$$\lim_{y\to y_0^+} F_X(y) = \lim_{n\to\infty} P_X(C_n) = P_X\left(\bigcap_{n=1}^{\infty} C_n\right) = P_X((-\infty, y_0]) = F_X(y_0).$$

□

Lemma 4.44.5 If two random variables X and Y are equal a.e., then they have the same cumulative distribution function.

Crux. Since $X = Y$ almost everywhere, the set where they differ has probability zero ($P(X \neq Y) = 0$). For any $y \in \mathbb{R}$:

$$P(X \le y) = P(\{X \le y\} \cap \{X = Y\}) + P(\{X \le y\} \cap \{X \neq Y\})$$

The second term is 0. On the set where $X = Y$, the condition $X \le y$ is identical to $Y \le y$. Therefore:

$$P(X \le y) = P(\{Y \le y\} \cap \{X = Y\}) = P(Y \le y)$$

where the last equality follows from adding back the zero-probability set $\{Y \le y\} \cap \{X \neq Y\}$. Thus, $F_X(y) = F_Y(y)$. □

Theorem 4.45. *(Skorokhod Representation Theorem) If a function $F : \mathbb{R} \to [0, 1]$ satisfies conditions (i)-(iii) of Proposition 4.44, then there is a random variable $X : [0, 1] \to \mathbb{R}$ defined on the probability space $([0, 1], \mathcal{B}, m_{[0,1]})$, such that $F_X = F$.*

Crux. Define $X^+(\omega) \triangleq \inf\{x : F(x) > \omega\}$ and $X^-(\omega) \triangleq \sup\{x : F(x) < \omega\}$. Since F is non-decreasing, the set $\{x : F(x) > \omega\}$ is strictly bounded below by the set $\{x : F(x) < \omega\}$. Therefore, $X^-(\omega) \le X^+(\omega)$ for all ω.

Part 1: CDF of X^- We establish the equivalence $X^-(\omega) \le y \iff \omega \le F(y)$. If $\omega \le F(y)$, then y bounds the set $\{x : F(x) < \omega\}$ from above, so $X^-(\omega) \le y$. Conversely, if $X^-(\omega) \le y$, the right-continuity of F guarantees $\omega \le F(y)$. Since $m_{[0,1]}$ is the uniform Lebesgue measure:

$$F_{X^-}(y) = m_{[0,1]}(\{\omega : X^-(\omega) \le y\}) = m_{[0,1]}(\{\omega : \omega \le F(y)\}) = F(y).$$

Part 2: Equivalence a.e. Suppose $\omega_1 < \omega_2$. By the monotonicity of F, $X^+(\omega_1) \le X^-(\omega_2)$. Thus, for any ω where $X^-(\omega) < X^+(\omega)$, the open intervals $(X^-(\omega), X^+(\omega))$ are strictly disjoint from one another. Since any collection of disjoint open intervals on the real line is at most countable (as each must contain a distinct rational number), the set $\{\omega : X^-(\omega) \ne X^+(\omega)\}$ is countable. A countable set has Lebesgue measure zero, so $X^- = X^+$ a.e.

By Lemma 4.44.5, $F_{X^+} = F_{X^-} = F$. Thus, both X^- and X^+ satisfy the theorem. $\square$

Theorem 4.46. *If P_X defined on $\mathbb{R}^n$ is absolutely continuous with density f_X, and $g : \mathbb{R}^n \to \mathbb{R}$ is integrable with respect to P_X, then*

$$\int_{\mathbb{R}^n} g \, \mathrm{d}P_X = \int_{\mathbb{R}^n} f_X(x) g(x) \, \mathrm{d}x.$$

Crux. We apply the standard measure-theoretic machine:

(1) **Indicators:** For $g = \mathbb{I}_A$ (with A Borel), the LHS is $P_X(A)$ and the RHS is $\int_A f_X(x)\,\mathrm{d}x$. These are exactly equal by the definition of an absolutely continuous density.

(2) **Simple Functions:** The equality extends to finite linear combinations of indicators via the linearity of the integral.

(3) **Non-negative Functions:** For $g \ge 0$, choose simple functions $s_n \nearrow g$. Applying the Monotone Convergence Theorem to both sides simultaneously preserves the equality for the limit.

(4) **General Functions:** For arbitrary integrable g, decompose $g = g^+ - g^-$, apply Step 3 to each part, and subtract.

$\square$

Corollary 4.47 If X is an n-dimensional random vector with absolutely continuous distribution P_X and density f_X, and $g : \mathbb{R}^n \to \mathbb{R}$ is Borel measurable and integrable, then

$$\int_\Omega g(X) \, \mathrm{d}P = \int_{\mathbb{R}^n} f_X(x) g(x) \, \mathrm{d}x.$$

NOTE: The authors silently generalize Theorem 4.41 from $\mathbb{R}$ to $\mathbb{R}^n$ here. This is mathematically legal because the "Standard Machine" proof of Theorem 4.41 relies entirely on abstract measurable sets (Borel sets), meaning the exact same proof logic holds for n-dimensional spaces.

Crux. By the n-dimensional generalization of Theorem 4.41, the LHS is equivalent to $\int_{\mathbb{R}^n} g \,\mathrm{d}P_X$. By Theorem 4.46, this immediately evaluates to the RHS. $\square$

Theorem 4.48. *If $g : \mathbb{R} \to \mathbb{R}$ is strictly increasing and differentiable, then the density of $g(X)$ is given by*

$$f_{g(X)}(y) = f_X(g^{-1}(y)) \frac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y).$$

NOTE: The text loosely says "increasing", but g must be **strictly** increasing to guarantee a well-defined inverse g^{-1}. If g were merely non-decreasing, it could have flat spots, destroying invertibility. Because g is differentiable (thus continuous) and monotonic, it is automatically Borel measurable, ensuring $g(X)$ is a valid random variable.

Crux. Using the CDF method and the fact that g^{-1} preserves inequalities (since g is strictly increasing):

$$F_{g(X)}(y) = P(g(X) \le y) = P(X \le g^{-1}(y)) = F_X(g^{-1}(y)).$$

Assuming P_X is absolutely continuous (has a density), we find the density of $g(X)$ by differentiating both sides with respect to y. Applying the chain rule to the RHS yields the result. $\square$

Remark 4.49 A similar result holds if g is strictly decreasing and differentiable. An argument similar as above gives

$$f_{g(X)}(y) = -f_X(g^{-1}(y)) \frac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y).$$

Definition: Mathematical Expectation If X is a random variable on $(\Omega, \mathcal{F}, P)$, its *mathematical expectation* $\mathbb{E}(X)$ is the abstract integral:

$$\mathbb{E}(X) \triangleq \int_\Omega X \,\mathrm{d}P.$$

In terms of the induced distribution P_X and density f_X (if absolutely continuous):

$$\mathbb{E}(X) = \int_{\mathbb{R}} x \,\mathrm{d}P_X(x) = \int_{\mathbb{R}} x f_X(x) \,\mathrm{d}x.$$

Definition: Complex Random Variables For real random variables X and Y, $Z \triangleq X + iY$ is a complex random variable. The Dominated Convergence Theorem, linearity of the integral, and the triangle inequality $\left|\int f \,\mathrm{d}m\right| \le \int |f| \,\mathrm{d}m$ naturally extend to the complex modulus $|Z|$.

Definition 4.52: Characteristic Function The *characteristic function* of X for $t \in \mathbb{R}$ is:

$$\varphi_X(t) \triangleq \mathbb{E}(e^{itX}) = \int_\Omega e^{itX} \,\mathrm{d}P.$$

In terms of P_X and density f_X:

$$\varphi_X(t) = \int_{\mathbb{R}} e^{itx}\,\mathrm{d}P_X(x) = \int_{\mathbb{R}} f_X(x)e^{itx}\,\mathrm{d}x.$$

Theorem 4.53. *The characteristic function φ_X satisfies:*

(1) $\varphi_X(0) = 1$ and $|\varphi_X(t)| \leq 1$ for all t.

(2) $\varphi_{aX+b}(t) = e^{itb}\varphi_X(at)$.

Crux. (1) Setting $t = 0$ yields $\mathbb{E}(e^0) = \mathbb{E}(1) = 1$. The bound follows directly from the integral triangle inequality and Euler's formula: $|\mathbb{E}(e^{itX})| \leq \mathbb{E}(|e^{itX}|) = \mathbb{E}(1) = 1$.

(2) By the linearity of expectation: $\mathbb{E}(e^{it(aX+b)}) = \mathbb{E}(e^{itb}e^{i(at)X}) = e^{itb}\mathbb{E}(e^{i(at)X}) = e^{itb}\varphi_X(at)$.

$\square$